Hasan Mohd. Hussain

Machine Learning Engineer | LLM & Generative AI Systems
Dallas, TX | (469) - 230 - 5131 | hasan16017@gmail.com | LinkedIn | **https://github.com/striderzz**
Work Authorization: F-1 STEM OPT (Valid through 2029) – No sponsorship required

PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building **agentic AI systems, LLM applications, and RAG pipelines** using LangGraph, LangChain, and vector databases. Skilled in embeddings and deep learning with PyTorch, and deploying scalable ML systems using FastAPI, Flask, and Docker. Experienced in developing **production-ready AI agents and intelligent retrieval pipelines.**

EDUCATION

University of Texas at Dallas — **Richardson, TX**
MS in Computer Engineering | GPA 3.75 / 4.0 — Aug 2024 – (Expected) Jun 2026
Relevant Coursework - DSA, Computer Architecture, Embedded Systems, Database Design, Machine Learning, Natural Language Processing

University of Mumbai — **Mumbai, India**
Bachelor's Degree in Computer Engineering | GPA 7.88 / 10.0 — Aug 2017 – Jul 2021
Relevant Coursework - DBMS, Data Structures, ML, NLP

SKILLS

AI / ML: Deep Learning, Transformers, LLMs, RAG, AI Agents, Embeddings, Semantic Search, Fine-Tuning, Feature Engineering, Model Evaluation
Generative AI & LLM Stacks: Azure OpenAI, Google Gemini, OpenAI API, LangChain, LangGraph, Vector Databases, ChromaDB
Frameworks & Libraries: PyTorch, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, NumPy, Pandas
Backend & Model Deployment: FastAPI, Flask, REST APIs, Model Serving, Docker, AWS Elastic Beanstalk, EC2, S3, EBS, CloudWatch
Databases: PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB
Programming Languages: Python, C++, C#, JavaScript
Tools & Systems: Git, GitHub, CodeCommit, CodePipeline, Linux/Unix, Shell Scripting, Postman

WORK EXPERIENCE

Mphasis — **New York, United States**
Machine Learning Engineer Intern — May 2025 - Aug 2025
- Implemented text embedding pipelines using **Azure Open AI** and integrated **vector databases** like **ChromaDB** to optimize retrieval performance by **30%** to enable LLM-based **Retrieval-Augmented Generation** (RAG) workflows
- Developed Python algorithms and backend logic for full-stack applications, delivering optimized code that **improved API response performance by 20%**.
- Worked with tools like FastAPI, Python, **AWS, Docker, LangChain,** Azure OpenAI, and Gemini APIs

Dot9 Games — **Mumbai, India**
Game Developer Intern — Apr 2024 - Aug 2024
- Implemented gameplay features using **C# and Unity**, translating design specifications into functional systems
- Collaborated with developers, artists, and designers in **Agile/Scrum workflows**, using **Git for version control** and iterative development

Nilee Games — **Mumbai, India**
Game Developer Intern — Nov 2022 - Mar 2023
- Developed a **2D physics-based game prototype in Unity using C#**, delivering the project within a two-week development cycle
- Applied **software engineering practices including debugging, iteration, and team collaboration** to improve gameplay systems and user engagement

TECHNICAL PROJECTS

RAG Document QA System with Gemini Integration
- Implemented a **Retrieval-Augmented Generation** pipeline using **LangChain, ChromaDB, and Hugging Face** embeddings for semantic search, document chunking, and PDF text retrieval.
- Integrated **Google Gemini LLM** to generate context-aware answers, boosting **query accuracy by 35%**.
- **Tools Used:** Python, **ChromaDB, Hugging Face, LangChain, Google Gemini**, PyPDF2

Deep Learning Image Classification (MLP vs CNN) – PyTorch
- Developed **MLP and CNN architectures** in PyTorch on MNIST and CIFAR-10, leveraging **SGD/Adam optimization, BatchNorm, Dropout**
- Conducted **hyperparam tuning** across 30+ experiments, optimizing learning rate, batch size, and regularization for improved generalization.
- **Achieved 99%+ accuracy (MNIST) and ~78% accuracy (CIFAR-10), demonstrating strong performance in supervised computer vision and spatial feature learning.**

LangGraph Agentic Nutrition System
- Built a **multi-agent nutrition AI system** using **LangGraph** and **GPT-4o-mini**, integrating **ReAct agents** and a **critic–revise feedback loop** with score-based routing.
- Implemented real-time **Flask SSE** streaming with live **Think-Act-Observe** reasoning traces per agent, producing a 7-day meal plan, supplement protocol, and safety-audited coaching report.

CERTIFICATIONS

DeepLearning.AI: Supervised Machine Learning (Regression & Classification), AI for Everyone
IBM: Machine Learning with Python, Building AI Agents and Agentic Workflows
Meta: Back-End Developer Professional Certificate